

Swapnil Alase

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Profile Summary

AI & ML Enthusiast / Senior Embedded Software Engineer at Forvia

Results-driven engineer with over 5 years of experience in automotive embedded systems and infotainment software. Currently transitioning into the field of Artificial Intelligence and Data Analytics through an MS in AI & ML from Scaler Neovarsity. Proficient in Python, SQL, and Statistics, with a growing focus on extracting insights from data and applying analytical approaches to solve real-world problems. Passionate about integrating data-driven thinking with engineering expertise to build smarter and more efficient systems.

Education

MS in Artificial Intelligence & Machine Learning — Scaler Neovarsity : affiliated with Woolf University (Ongoing)

Industry-focused curriculum covering SQL, Python, Statistics, Data Analytics, and additional advanced topics as part of an ongoing MS in AI & ML program.

PG-Diploma in Embedded Systems Design — CDAC, Pune (2020)

Bachelor of Engineering (Electronics & Telecommunication) — SCOE, University of Pune (2017)

Key Technical Skills

Programming: Python, C, C++, Embedded C, SQL

Libraries/Tools: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook

Concepts: Data Analytics, Statistics, Hypothesis Testing, Correlation, Data Visualization, Exploratory Data analysis (EDA)

Embedded Expertise: Bluetooth, Infotainment Systems, Git, Cybersecurity, Jenkins, Linux

AI & Data Analytics Projects

Target — Order Data Analytics (SQL)

Analyzed 100K+ e-commerce orders (2016–2018) to evaluate order statuses, payments, freight, and customer demographics. Generated actionable insights that improved pricing strategy, shipping efficiency, and customer satisfaction.

Skills: SQL, Data Cleaning, Exploratory Data Analysis

Walmart — Confidence Interval & CLT Study (Python)

Explored 100M+ customer transactions to analyze gender-based, age-based, marital status based purchasing and trends. Strengthened marketing strategies with statistical validation using CLT and confidence intervals.

Skills: Python, Probability, Statistics, Hypothesis Testing

AeroFit — Customer Profiling through Descriptive Analytics

Performed customer segmentation and descriptive analysis to support targeted marketing strategies and data-driven business decisions.

Skills: Statistics, Data Visualization, Probability Distributions

Netflix — Global Data Exploration & Visualization

Analyzed Netflix's global dataset to identify regional growth opportunities and content trends. Delivered actionable insights for improving content strategy.

Skills: Data Visualization, EDA, Insight Generation

Professional Experience

- **Forvia (formerly Faurecia) — Pune, India**

Senior Software Engineer (April 2024 – Present)

- Led the CEA team (3 members) to establish Bluetooth communication between the car's radio and mobile app using the CEA protocol.
- Integrated cybersecurity mechanisms ensuring secure data exchange between infotainment and IVI modules.
- Collaborated with cross-functional teams to meet Stellantis infotainment standards.

Software Engineer (March 2021 – April 2024)

- Developed and validated infotainment modules for Volvo, Renault & Mack trucks.
- Implemented voice recognition contexts and middleware test benches (FC7100, NIS).

Graduate Engineering Trainee (March 2020 – March 2021)

- Contributed to feature validation and testing for infotainment systems.

- **Axonet Emsys Pvt. Ltd — Pune, India**

Embedded Intern (Dec 2017 – May 2018)

- Developed IoT-based gateways using PIC microcontrollers; implemented Modbus, I²C, and UART communication.

Certifications

- Scrum Fundamentals Certified (SFC)
- What is Generative AI? (LinkedIn Learning)
- Generative AI: The Evolution of Thoughtful Search (LinkedIn Learning)

Publications

IoT Based Digital Signage Board using Raspberry Pi 3 — IRJET, May 2017